

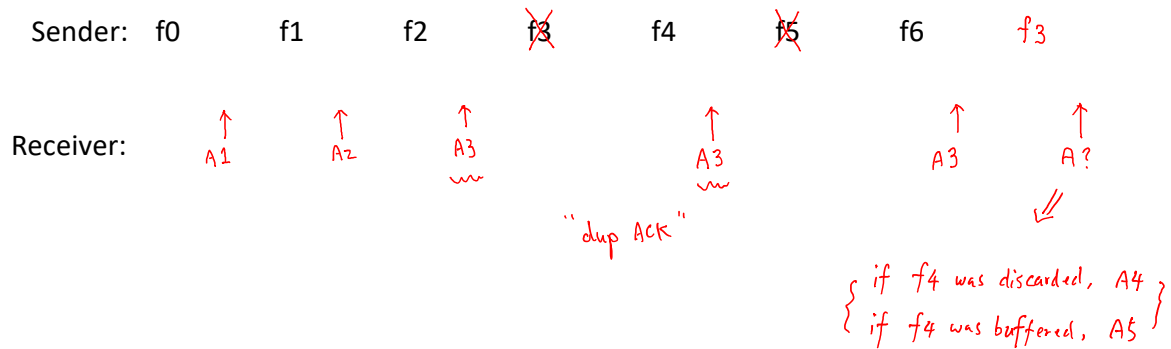
SN:

- SN of data frame : position in the sequence
- SN carried in control frames :
regarding whether or which data frames in the sequence
have been received so far

"cumulative acknowledgment"

E.g. $\text{ACK}(20) \Rightarrow$ all data frames with $\text{SN} < 20$
have been received correctly
 $\Rightarrow$ the next data frame that receiver
is waiting for is #20

CPR E 4890 -- D.Q.



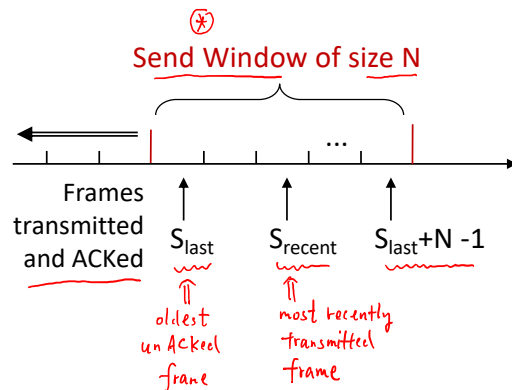
CPR E 4890 -- D.Q.

1. Go-Back-N ARQ (GBN)

ACK only , No NAK

- GBN allows the transmitter to keep sending data frames while waiting for acknowledgment from the receiver

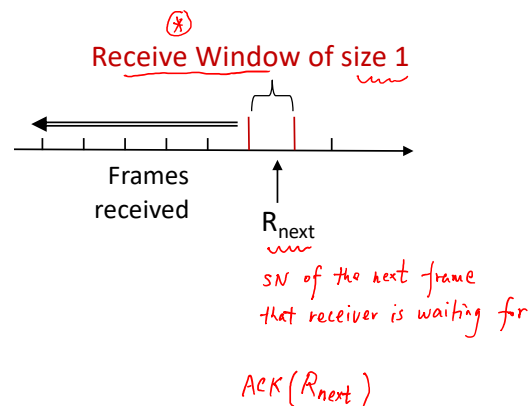
- At the transmitter side:
Allow up to N
outstanding frames



CPR E 4890 -- D.Q.

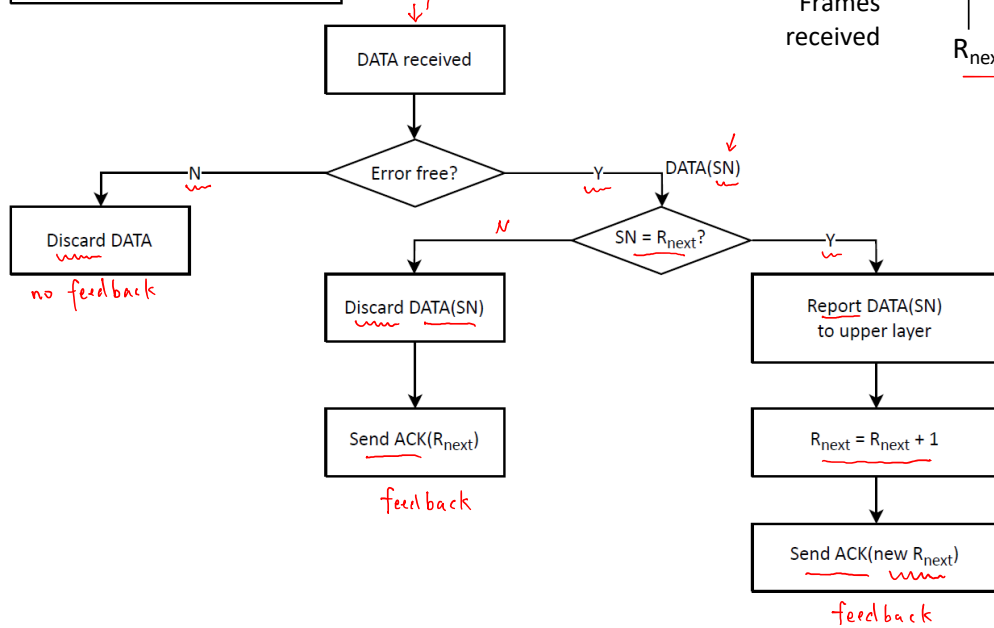
- GBN allows the transmitter to keep sending data frames while waiting for acknowledgment from the receiver

- At the receiver side:
Receive window of size 1



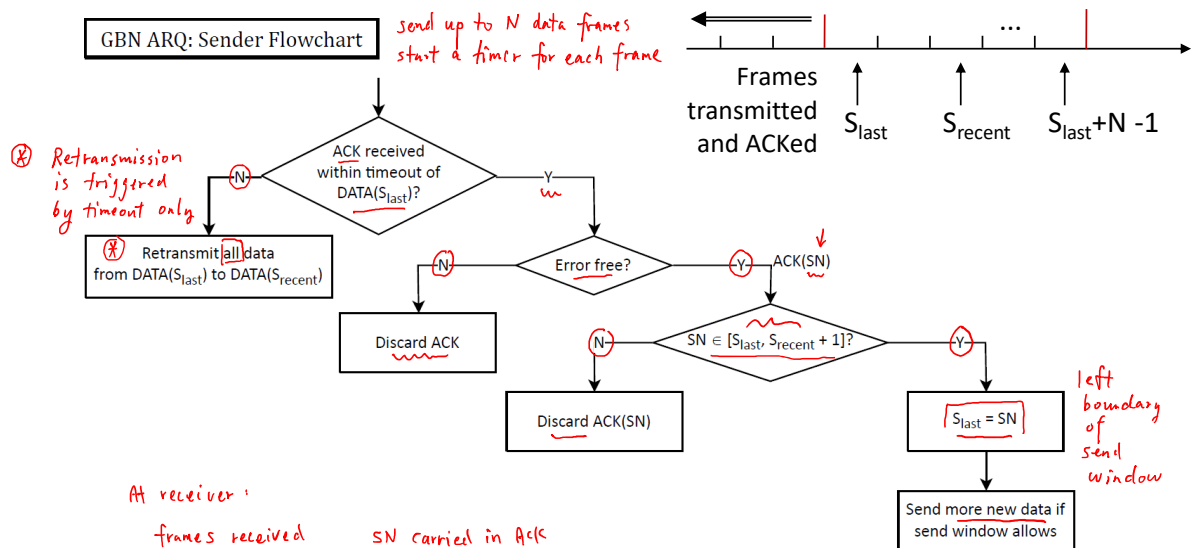
CPR E 4890 -- D.Q.

GBN ARQ: Receiver Flowchart



CPR E 4890 -- D.Q.

GBN ARQ: Sender Flowchart



At receiver:

frames received

SN carried in ACK

S_{last}
 $\rightarrow S_{last}, \dots, S_{recent}$
 $S_{last}, S_{last} + 1, \dots, S_{recent}$

$S_{last} + 1$
 $S_{recent} + 1$
 S_{last}

CPR E 4890 -- D.Q.

